

Owed gate implementations – § 11.4.227(A) deferral registry

Revision: 3 **Last modified:** 2026-08-20T15:37:04Z **Description:** The tracked work items every row of `scripts/gates/gate_ledger_deferrals.tsv` points at. **Authority:** §11.4.227(A) (named-gate ledger + monotone-decrease ratchet), §11.4.197 (started work reaches a terminal state), §11.4.6 (no-guessing).

Why this document exists

§11.4.227(A) requires every `CM-*` gate named in the governance corpus to be **either** implemented in an executable gate site **or** covered by a **registered deferral pointing at a tracked item**. Silent gate debt is forbidden, and so is deleting a name to lower the count.

64 of this document's 76 `OWED-GATE-NNN` entries are still genuinely owed — confirmed 2026-08-20T15:32:40Z by diffing every `OWED-GATE-NNN` entry's backtick-quoted gate name (Sections A

- C, numeric ids only — the non-numeric `OWED-GATE-ADV-F*` findings in Section B.1 are review findings, not gates, and are excluded) against the first column of `scripts/gates/gate_ledger_deferrals.tsv` (72 rows total; a control-needle overlap check confirmed the diff instrument is not blind, §11.4.201(7)(b)). The remaining 12 entries already resolve to a landed gate-site file under `constitution/scripts/gates/` — they are IMPLEMENTED, not yet pruned from this list, and their discharge is a *different* stream's call, backed by a passing §1.1 mutation test (§11.4.227(A) Acceptance Criterion 5). **This count is a live snapshot, not a fixed figure** — the TSV is under active concurrent implementation and the number moves; re-derive it by re-running the same diff rather than trusting this sentence after the fact. *(Historical note: an earlier draft of this sentence read "66" — no interpretation checked against evidence at the*

time this note was written [Section A entry count, this document's total `OWED-GATE-NNN` count, the TSV's row count, or the ledger's own live `unimplemented=` figure per §11.4.227(A), which reads 420 and matches the checked-in `gate_ledger_baseline.txt`] equalled 66 under any of those readings, so the figure has been replaced with the one above rather than guessed at, §11.4.6.) Each still-owed gate needs a genuine per-domain harness (a build-reproducibility comparison, a SLSA attestation verifier, a contract broker, a diagram-accuracy oracle, ...). **Implementing them as stubs that cannot FAIL would be the exact §11.4 bluff this corpus forbids** — a gate that cannot refuse is decoration, and per §11.4.115(F) a guard never observed FAILing on a genuinely-broken artifact is unvalidated instrumentation that mints nothing. So they are registered as **owed debt**, visibly counted and monotone-decreasing, rather than faked green.

Honest boundary (§11.4.6): registering a deferral proves the debt is *tracked*, not that the gate *works*. Each item is discharged only when its executable site lands with a paired §1.1 mutation observed to make it FAIL.

Acceptance criteria (apply to every `OWED-GATE-*` item)

An item is discharged when **all** hold:

1. An executable gate site exists under `scripts/` carrying the gate's literal name structurally (§11.4.201(7)(a) — a prose carrier never counts).
2. A paired §1.1 mutation exists and was **observed** to make the gate FAIL before the gate is trusted (§11.4.115(F) observation-before-trust).
3. Golden-good **and** golden-bad fixtures are wired, plus a negative control where a false-positive refusal is possible (§11.4.107(10), §11.4.201(1)).
4. Any absence the gate reports is control-needle-proven (§11.4.201(7)(b)).
5. The gate's row moves out of `gate_ledger_deferrals.tsv`, and the ledger baseline in `scripts/gates/gate_ledger_baseline.txt` ratchets **down**, never up (§11.4.227(A)).

A. Owed gate implementations

§ . .

• OWED-GATE-053 — CM-TEST-TYPE-BREADTH-SEVEN-CANONICAL

- **Must assert:** asserts a consuming project's suite covers all seven canonical test types where applicable with honest SKIP-with-reason where not

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- **Blocked on:** needs the canonical type list and per-type detector as consumer DATA

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• OWED-GATE-012 — CM-CLOSURE-VOCAB-TYPE-AWARE

- **Must assert:** asserts every closed item Status matches its Type - Bug to Fixed, Feature to Implemented, Task to Completed

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- **Blocked on:** the Issues/Fixed tracker DATA is consumer-owned and absent from this project-agnostic repo

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• OWED-GATE-071 — CM-ITEM-SLICED-SMALL-AND-TESTABLE

- **Must assert:** asserts every item carries acceptance criteria whose satisfaction is decidable from that item's own evidence and is a vertical slice - a thin end-to-end capability a user can observe; a horizontal-layer item, or one whose criteria depend on a sibling item landing, FAILs naming the item
- **Blocked on:** the workable-items tracker DATA is consumer-owned and absent from this project-agnostic repo, the same blocker class as OWED-GATE-012
- **Paired 1.1 mutation:** re-shape one item into a horizontal layer whose criteria cannot be met alone and the gate MUST FAIL; golden-FALSE per 11.4.201(1) - a legitimately-dependent item that is nonetheless independently testable once its dependency is met MUST NOT fire it

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• **OWED-GATE-075** — CM-SUITE-SHAPE-DECLARED-AND-PUSHED-DOWN

- **Must assert:** asserts the project declares its chosen suite shape together with the risk-location justification that selected it, and that a high-level test failing with no lower-level counterpart opens a tracked 11.4.197 item to add the lower-level test
- **Blocked on:** needs a consumer suite-shape declaration plus a test-level classifier able to tell high-level from low-level tests, neither adopted
- **Paired 1.1 mutation:** remove the shape declaration, or record a high-level-only failure with no tracked lower-level follow-up, and the gate MUST FAIL; golden-FALSE per 11.4.201(1) - a declared trophy shape on a genuinely I/O-heavy system MUST NOT fire it merely for having few unit tests

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• **OWED-GATE-069** — CM-DEPENDENCY-POINTER-PREFLIGHT-FAIL-CLOSED

- **Must assert:** asserts the preflight resolves every submodule pointer, verifies it matches the intended ref, fails closed on a missing checkout, and for the truly load-bearing components asserts a runtime signature on the flashed or deployed artifact - closing all three mess classes stale, absent-uninitialised and unfetchable
- **Blocked on:** separating the unfetchable class from the merely-absent class requires probing configured remotes, and the hermetic pre-build gate run has no guaranteed network reach
- **Paired 1.1 mutation:** de-initialise one submodule and the gate MUST FAIL; point one gitlink at a commit no configured remote serves and it MUST FAIL; golden-FALSE - a fully-initialised tree at its intended refs MUST NOT fire it

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• **OWED-GATE-013** — CM-COMMIT-PUSH-ALWAYS-UNBLOCKED

- **Must assert:** asserts a failing validation yields a clear per-check report and a documented remediation path, never an opaque hung or rejected push

- **Blocked on:** the consumer binds its own commit/push script path as DATA per 11.4.35

- **OWED-GATE-019** — CM-DEDICATED-HOOK-VALIDATION-SCRIPT

- **Must assert:** asserts hook validations run as an explicit named stage of a dedicated idempotent commit/push script
- **Blocked on:** the consumer binds the script path as DATA per 11.4.35

- **OWED-GATE-029** — CM-HOOKS-NEVER-BLOCK-PUSH

- **Must assert:** asserts core.hooksPath does not gate routine commit/push while the hook file itself stays preserved and unmodified
- **Blocked on:** the consumer binds its repo and hook paths as DATA

- **OWED-GATE-038** — CM-NO-GATE-LOST-ON-HOOK-DISCONNECT

- **Must assert:** asserts every check a disconnected hook performed still executes at a named script stage or release gate
- **Blocked on:** needs the consumer-owned stage list and deferral registry

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- **OWED-GATE-045** — CM-QA-DEPLOY-READINESS-GATE

- **Must assert:** asserts no manual-QA hand-off occurs without a machine-written PASS verdict whose artifact fingerprint equals the candidate
- **Blocked on:** requires a verdict store keyed on artifact fingerprint

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- **OWED-GATE-054** — CM-TRANSLATION-CONTEXT-SPIRIT-REVIEW

- **Must assert:** asserts each localized artifact carries an independent review verdict with per-dimension rationale across the six dimensions
- **Blocked on:** requires a translation-review verdict store and a reviewer substrate

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• **OWED-GATE-036** — CM-MANUAL-QA-FINDS-NOTHING-NEW

- **Must assert:** asserts a manual QA pass discovered zero previously-unknown defects
- **Blocked on:** requires a discovery-channel ledger recording each defect's origin channel

• **OWED-GATE-046** — CM-QA-IS-THE-DISCOVERER

- **Must assert:** asserts every out-of-band discovery triggers a coverage-escape audit and registers a new automated check that would have caught

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it

- **Blocked on:** requires the discovery-channel ledger

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• **OWED-GATE-015** — CM-CRITICAL-INVARIANT-FAILURE-PATH-DOD

- **Must assert:** asserts every change touching money, safety, availability or integrity carries explicit failure-path scenarios as passing tests mapped to verdict pairs

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- **Blocked on:** needs a consumer criticality classification and wiring into the 11.4.146(D3) status-custody chain

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• **OWED-GATE-034** — CM-LEAST-PRIVILEGE-CAPABILITY-NOT-INSTRUCTION

- **Must assert:** asserts producer-versus-gate separation is enforced by capability - filesystem scope, branch protection, store isolation - never by prompt or policy text
- **Blocked on:** requires CI permission-model introspection

• **OWED-GATE-041** — CM-PRODUCER-NOT-VERIFIER-SCOPE-SPLIT

- **Must assert:** asserts producer, oracle, gate and verifier resolve to distinct actors with no collapse involving the producer
- **Blocked on:** requires an actor-scope registry

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• **OWED-GATE-032** — CM-ILLEGAL-STATE-LADDER-PREFERENCE

- **Must assert:** asserts each load-bearing invariant lands at the strongest rung the toolchain supports - types over API-shape over lint over property-test over runtime-assert

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- **Blocked on:** needs per-language toolchain capability detection and a rung classifier

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• **OWED-GATE-005** — CM-BISECTION-BEFORE-BLAME

- **Must assert:** asserts regression localisation used bisection from the last known-good tag with a monotone-observable precheck rather than blame-log walking
- **Blocked on:** needs a per-investigation evidence record format, none adopted yet

• **OWED-GATE-068** — CM-DELTA-DEBUG-MINIMAL-CASE

- **Must assert:** asserts an investigation of a non-monotone or non-commit-discriminated failure records the narrowed axis and its minimal failing case, and that this case is the 11.4.115 RED test's input; a fix designed against an un-narrowed failing input FAILs
- **Blocked on:** needs the same per-investigation evidence record format still owed by OWED-GATE-005, none adopted yet

11 4 2 4 3

- **Paired 1.1 mutation:** strip the minimal-case record so the fix is designed against the full unreduced input and the gate MUST FAIL; golden-FALSE per 11.4.201(1) - a genuine monotone commit-range regression localised by git bisect alone MUST NOT fire it

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• **OWED-GATE-011** — CM-CHARACTERIZATION-BEFORE-LEGACY-MODIFICATION

- **Must assert:** asserts behaviour-changing edits to legacy code are preceded by a landed green characterization baseline

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- **Blocked on:** needs a legacy-classification rule and a baseline-precedence record

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• **OWED-GATE-009** — CM-CAN-I-DEPLOY-GATE-WIRED

- **Must assert:** asserts deploy is blocked unless the provider-by-consumer compatibility matrix is all-green, with a missing contract treated as a first-class refusal
- **Blocked on:** requires a contract broker holding deployed-version state

• **OWED-GATE-014** — CM-CONTRACT-TESTS-BOTH-SIDES

- **Must assert:** asserts each changed cross-boundary carries consumer-published and provider-verified contract tests on both sides
- **Blocked on:** requires a contract broker and a boundary inventory

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• **OWED-GATE-040** — CM-ORACLE-STRATEGY-NAMED-AND-INDEPENDENT

- **Must assert:** asserts each test names its oracle strategy from the closed seven-member set and that the oracle is structurally independent of the code under test
- **Blocked on:** requires an oracle-annotation convention and parser, none adopted

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• **OWED-GATE-007** — CM-BUILD-HERMETICITY

- **Must assert:** asserts no undeclared build inputs - every dependency version-pinned and lockfile-hashed, no unpinned network reach, no host-env leakage
- **Blocked on:** requires a sandboxed no-network build harness that does not exist here

• **OWED-GATE-008** — CM-BUILD-REPRODUCIBILITY

- **Must assert:** asserts two independent builds of identical source yield byte-identical artifacts or a diff-set fully accounted for by declared non-determinisms
- **Blocked on:** requires building real artifacts twice on conformant hosts - this repo has no build pipeline

• **OWED-GATE-050** — CM-SLSA-L2-PROVENANCE-VERIFIED

- **Must assert:** asserts signed SLSA Build Level 2 provenance - source URI, commit hash, builder identity, invocation, artifact hash - verifies against the artifact
- **Blocked on:** requires a hosted tamper-resistant build platform emitting attestations

• **OWED-GATE-052** — CM-SUPPLY-CHAIN-DEPS-PROVENANCE-ATTESTED

- **Must assert:** asserts dependencies are vendored hash-verified,
1 1 4 2 4 7 provenance-attested at source, or mirrored through a hash-pinning registry
- **Blocked on:** requires a dependency inventory and an attestation verifier

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• **OWED-GATE-033** — CM-LAYER-MOVE-COMPLETENESS-CHECK

- **Must assert:** asserts a boundary move leaves zero residual references across all seven layers or an explicit tracked compatibility window per residual
- 1 1 4 2 4 8 ◦ **Blocked on:** requires the consumer layer-taxonomy audit script and a layer inventory

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• **OWED-GATE-025** — CM-FLAKY-TEST-QUARANTINE-WIRED

- **Must assert:** asserts a test-history aggregator detects flakes, moves them to quarantine and opens a deadlined stabilisation item
- **Blocked on:** requires multi-run test-history storage

• **OWED-GATE-044** — CM-PROTECTED-REGRESSION-SPEC-GATE

- **Must assert:** asserts tests tagged PROTECTED-SPEC require the designated CODEOWNERS reviewer before merge

1.1 4 2.4.9

- **Blocked on:** requires forge branch-protection and CODEOWNERS introspection

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• **OWED-GATE-026** — CM-FLIGHT-RECORDER-CAPTURED

- **Must assert:** asserts producer inputs, oracle verdict, gate decision and verifier audit are captured as a durable replayable artifact
- **Blocked on:** requires an event-stream recorder

• **OWED-GATE-048** — CM-ROLE-SEPARATION-DECLARED

- **Must assert:** asserts each load-bearing quality mechanism declares its four roles with no forbidden collapse
- **Blocked on:** requires a role-declaration convention

1.1 4 2.5.0

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• **OWED-GATE-028** — CM-HEURISTIC-TOWER-TRIGGERS-PRIMITIVE-DEBUG

- **Must assert:** asserts a stack of two or more compensating heuristics triggers a primitive systematic-debug instead of adding layer N+1
- **Blocked on:** needs a tower detector that distinguishes accreted mitigation from deliberate defense-in-depth

1.1 4 2.5.1

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• **OWED-GATE-037** — CM-NO-BYTE-IDENTICAL-FORK

- **Must assert:** asserts no two near-identical trees exist differing only by configuration values
- **Blocked on:** requires a cross-tree similarity analyzer plus consumer-owned threshold DATA

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- **OWED-GATE-017** — CM-DANGEROUS-COMBINATION-FAIL-CLOSED
 - **Must assert:** asserts every path combining two or more dangerous capabilities refuses on any unverifiable precondition and never fails open
 - **Blocked on:** needs a capability-taxonomy static analyzer, none exists

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- **OWED-GATE-030** — CM-IDEMPOTENCY-CHAOS-VERIFIED
 - **Must assert:** asserts simultaneous-retry, mid-operation-crash and retry-after-apparent-success chaos cases all pass
 - **Blocked on:** requires a live database plus a fault-injection harness
- **OWED-GATE-031** — CM-IDEMPOTENCY-DB-LEVEL-UNIQUE-GUARD
 - **Must assert:** asserts every retryable operation carries an idempotency key guarded by a database UNIQUE constraint rather than an application-level check-then-insert
 - **Blocked on:** requires schema introspection of a real database

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- **OWED-GATE-006** — CM-BOOT-TIME-INVARIANT-ASSERTED
 - **Must assert:** asserts each service asserts env, config, dependency, credential, schema and host invariants at boot and exits non-zero rather than serving
 - **Blocked on:** this repo ships no services, so it needs a consumer service inventory as DATA
- **OWED-GATE-010** — CM-CAPABILITY-MATRIX-PRESENT-AND-IN-SYNC
 - **Must assert:** asserts the capability matrix declares boot invariants plus offered and required capabilities and stays fingerprint-fresh
 - **Blocked on:** needs a service inventory and an emitted-endpoint extractor

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- **OWED-GATE-024** — CM-FAQ-FROM-REAL-QUESTIONS
 - **Must assert:** asserts a FAQ exists per component, derived from real operator, QA and end-user questions rather than invented
 - **Blocked on:** needs a component inventory and a captured question corpus
- **OWED-GATE-027** — CM-GUIDE-PER-USER-WORKFLOW
 - **Must assert:** asserts a task-oriented guide exists for every user-facing workflow
 - **Blocked on:** needs a workflow inventory as consumer DATA
- **OWED-GATE-055** — CM-USER-MANUAL-PER-COMPONENT
 - **Must assert:** asserts a complete user manual exists per component, service and product
 - **Blocked on:** needs a component inventory as consumer DATA

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- **OWED-GATE-001** — CM-ARCHITECTURE-DIAGRAM-PRESENT-ACCURATE
 - **Must assert:** asserts a system-architecture diagram exists in an open diffable format, renders non-blank, and matches the live deployed topology
 - **Blocked on:** needs a diagram renderer plus a topology-extraction oracle able to prove diagram-vs-reality accuracy - neither exists in-repo
- **OWED-GATE-018** — CM-DATAFLOW-DIAGRAM-PER-SERVICE
 - **Must assert:** asserts a data-flow diagram per service, accurate against real data movement and embedded at its point of use
 - **Blocked on:** same renderer and accuracy-oracle gap as the architecture diagram gate
- **OWED-GATE-020** — CM-DIAGRAM-EMBEDDED-NOT-ORPHAN
 - **Must assert:** asserts every diagram is referenced from a README-reachable document rather than living in an orphan gallery
 - **Blocked on:** depends on the diagram inventory the other 11.4.258 gates establish

- **OWED-GATE-049** — CM-SEQUENCE-DIAGRAM-PER-CRITICAL-WORKFLOW
 - **Must assert:** asserts a sequence diagram per critical user workflow and per critical cross-component protocol
 - **Blocked on:** needs a workflow inventory and a renderer
- **OWED-GATE-051** — CM-STATEMACHINE-DIAGRAM-PER-STATEFUL-ENTITY
 - **Must assert:** asserts a state-machine diagram per stateful entity matching that entity's real transitions
 - **Blocked on:** needs an entity inventory and a transition extractor

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- **OWED-GATE-002** — CM-BADGE-CLOSED-COLOR-VOCABULARY
 - **Must assert:** asserts every README badge colour is drawn from the closed set green/amber/red/gray with every gray carrying an explicit reason
 - **Blocked on:** needs a badge-row parser and a consumer-owned badge set defined as DATA per 11.4.35
- **OWED-GATE-003** — CM-BADGE-MACHINE-DERIVED-SOURCE
 - **Must assert:** asserts each badge value traces to a machine-derived provenance source and is never hand-typed
 - **Blocked on:** requires a per-badge provenance computer wired to build, test, coverage and security sources that do not yet exist
- **OWED-GATE-004** — CM-BADGE-SELF-VALIDATED
 - **Must assert:** asserts the badge computer passes golden-good, golden-bad and negative-control fixtures per 11.4.107(10)
 - **Blocked on:** depends on the badge computer landing first
- **OWED-GATE-042** — CM-PRODUCTION-READINESS-GAUGE
 - **Must assert:** asserts the composite readiness gauge reads green only when every invariant and every blocker is clear
 - **Blocked on:** depends on the readiness tracker and the badge computer

• **OWED-GATE-047** — CM-README-BADGE-ROW-AT-TOP

- **Must assert:** asserts the badge row sits directly below the H1 and above the introduction

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- **Blocked on:** the consumer badge set is undefined and a README structural parser is needed

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• **OWED-GATE-066** — CM-CORRELATION-ID-THREADED-ACROSS-BOUNDARIES

- **Must assert:** asserts a declared request or trace id is emitted by every signal and survives every async hop - queue, worker and scheduled sweep; a hop that drops it FAILs
- **Blocked on:** this repo ships no running services emitting signals across async hops, so it needs a consumer service and signal inventory as DATA
- **Paired 1.1 mutation:** drop the correlation id at the queue boundary and the gate MUST FAIL; golden-FALSE per 11.4.201(1) - a surface with genuinely no async hop and no crash-reporting channel, honestly 11.4.3 SKIP-with-reason, MUST NOT fire it

• **OWED-GATE-067** — CM-CRASH-FREE-RELEASE-GATE

- **Must assert:** asserts a crash-free users or sessions gate exists with a declared, locally-calibrated threshold, and that a crash signature new at the release boundary blocks regardless of volume
- **Blocked on:** requires a crash-reporting backend and a calibrated per-project threshold, neither available in this project-agnostic repo; the anchor mandates that the gate exists and that its threshold is declared locally, never a constitutional numeric value
- **Paired 1.1 mutation:** admit a release carrying a new crash signature at low volume and the gate MUST FAIL; golden-FALSE per 11.4.201(1) - a surface with no crash-reporting channel, honestly 11.4.3 SKIP-with-reason, MUST NOT fire it

• **OWED-GATE-016** — CM-CUTTING-EDGE-POSTURE-ALWAYS-ON

- **Must assert:** asserts the ten production-readiness invariants are evaluated for every change

- **Blocked on:** depends on the production-readiness tracker landing first

• **OWED-GATE-043** — CM-PRODUCTION-READINESS-TRACKER-PRESENT

- **Must assert:** asserts the per-component by invariant by evidence-source readiness tracker exists and is fresh
- **Blocked on:** needs a component inventory as consumer DATA

• **OWED-GATE-058** — CM-ZERO-NASTY-SURPRISES-AUDIT

- **Must assert:** asserts no known-broken code ships, no gate is silenced and no test is disabled without a tracked follow-up
- **Blocked on:** depends on the findings sweep and a tracker

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• **OWED-GATE-021** — CM-EVERY-FINDING-CLOSED-OR-TRACKED

- **Must assert:** asserts each audit-sweep finding is either closed with defect-layer evidence or carries a tracked item plus a regression guard
- **Blocked on:** depends on the findings sweep and a tracker

• **OWED-GATE-056** — CM-ZERO-FINDINGS-AUDIT-SWEEP

- **Must assert:** asserts the ten-class findings sweep runs and emits a machine-readable findings ledger
- **Blocked on:** requires the sweep script plus per-class golden-good and golden-bad fixtures

• **OWED-GATE-057** — CM-ZERO-FINDINGS-MONOTONE-RATCHET

- **Must assert:** asserts per-class and total findings counts never increase
- **Blocked on:** depends on the sweep and an operator-owned brownfield baseline decision per 11.4.66

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• **OWED-GATE-022** — CM-EVIDENCE-ANALYZER-SELF-VALIDATED

- **Must assert:** asserts every evidence analyzer passes golden-good, golden-bad and negative-control fixtures

- **Blocked on:** requires the analyzer fleet to exist before it can be validated

- **OWED-GATE-023** — CM-EVIDENCE-LAYER-MATCH-108

- **Must assert:** asserts each 11.4.108 layer PASS cites evidence produced AT that layer, rejecting wrong-layer echo
- **Blocked on:** requires a four-layer verdict store

- **OWED-GATE-035** — CM-MACHINE-EVIDENCE-AT-EVERY-GATE

- **Must assert:** asserts every works, passes or verified claim cites machine-created evidence by path, sha256 and timestamp
- **Blocked on:** requires a content-addressed evidence store

- **OWED-GATE-039** — CM-NO-NARRATIVE-ONLY-PASS

- **Must assert:** asserts no PASS rests on prose, operator eyeballing or an absence-of-error observation
- **Blocked on:** requires the evidence store to mechanically distinguish narrative from artifact

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- **OWED-GATE-063** — CM-BUILD-ONCE-PROMOTE-DIGEST

- **Must assert:** asserts the pipeline's build step is invoked at most once per release candidate and that every post-build environment stage references a pinned content-addressed digest and NOT a rebuild target; a stage that re-invokes the build FAILs
- **Blocked on:** this project-agnostic repo ships no deployment pipeline, so the gate needs a consumer pipeline definition as DATA per 11.4.35
- **Paired 1.1 mutation:** add a second per-environment build invocation and the gate MUST FAIL; replace a promoted digest reference with a mutable tag and it MUST FAIL; golden-FALSE per 11.4.201(1) - a single build followed by N digest-referencing promotion stages MUST NOT fire it

- **OWED-GATE-070** — CM-DEPLOY-CONFIG-INJECTED-NOT-BAKED

- **Must assert:** asserts no environment-specific endpoint, secret or flag is materialised into the artifact at build time; a per-environment build argument that changes artifact bytes FAILs

- **Blocked on:** proving byte-equality across two environment configurations requires a real build pipeline able to produce artifacts twice, which this repo does not have
- **Paired 1.1 mutation:** bake an environment endpoint into the build and the gate MUST FAIL; golden-FALSE per 11.4.201(1) - a single build followed by N digest-referencing promotion stages MUST NOT fire it

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• OWED-GATE-064 — CM-CANARY-ANALYSIS-INCLUDES-BUSINESS-METRIC

- **Must assert:** asserts the declared metric set contains at least one business or domain outcome metric and not exclusively infrastructure metrics; an infra-only analysis on a 11.4.239 critical-invariant surface FAILs
- **Blocked on:** depends on the progressive-delivery strategy declaration landing first, since the metric set is a field of that declaration
- **Paired 1.1 mutation:** strip every business metric from the analysis set leaving only error-rate and latency and the gate MUST FAIL; golden-FALSE per 11.4.201(1) - a genuinely non-routable artifact carrying an honest SKIP-with-reason MUST NOT fire it

• OWED-GATE-073 — CM-PROGRESSIVE-DELIVERY-STRATEGY-DECLARED

- **Must assert:** asserts a user-facing release either declares a bounded-blast-radius strategy with an automated analysis step, or carries an honest 11.4.3 SKIP-with-reason naming the absent traffic surface; an ungated full cutover with neither FAILs
- **Blocked on:** needs a consumer release-strategy declaration format and a traffic-surface inventory, neither adopted
- **Paired 1.1 mutation:** replace the automated abort with a manual approval step and the gate MUST FAIL; golden-FALSE per 11.4.201(1) - a genuinely non-routable artifact carrying an honest SKIP-with-reason MUST NOT fire it

- **OWED-GATE-065** — CM-CLAIM-REALITY-LEDGER-COMPLETE

- **Must assert:** asserts every advertised capability on every declared advertised surface resolves to exactly one ledger row; an advertised capability with no row FAILs naming the capability and the surface it was advertised on
- **Blocked on:** needs a consumer advertised-surface inventory plus a capability extractor, neither of which exists in-repo
- **Paired 1.1 mutation:** add a README capability with no ledger row and the completeness gate MUST FAIL; golden-FALSE per 11.4.201(1) - a fully-populated ledger whose every row has a fresh candidate-fingerprinted PASS verdict MUST NOT fire it

- **OWED-GATE-072** — CM-LEDGER-ROW-TYPED-FROM-CLOSED-VOCABULARY

- **Must assert:** asserts every ledger row carries a bluff type drawn from the seven-member closed set green-but-broken, coverage-theater, rubber-stamp-verified, stubbed-core, doc-vs-code-drift, config-present-but-unwired and byte-identical-fork; an ad-hoc type FAILs
- **Blocked on:** depends on the claim-vs-reality ledger schema landing first, there is no ledger to type-check yet
- **Paired 1.1 mutation:** retype a row to an invented type and the vocabulary gate MUST FAIL; golden-FALSE per 11.4.201(1) - a fully-populated ledger whose every row has a fresh candidate-fingerprinted PASS verdict MUST NOT fire it

- **OWED-GATE-076** — CM-UNCHALLENGED-CAPABILITY-BLOCKS-RELEASE

- **Must assert:** asserts a ledger row whose challenge is absent, never executed, or has no verdict for the release-candidate artifact fingerprint blocks the release exactly as a FAIL does
- **Blocked on:** depends on the claim-vs-reality ledger of OWED-GATE-065 plus a verdict store keyed by candidate fingerprint

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- **Paired 1.1 mutation:** blank a row's challenge reference and the blocking gate MUST FAIL; golden-FALSE per 11.4.201(1) - a fully-populated ledger whose every row has a fresh candidate-fingerprinted PASS verdict MUST NOT fire it

§ . .

• **OWED-GATE-062** — CM-ATTEMPT-RECORD-SHARED-AND-CONSULTED

- **Must assert:** asserts every multi-attempt effort writes each attempt and its outcome to the declared shared record, and that the record is read before the next attempt is dispatched; a second attempt duplicating a recorded-failed approach with no recorded material difference FAILs
- **Blocked on:** no shared attempt-record store is declared or adopted in-repo, and the consumer binds its store path as DATA per 11.4.35
- **Paired 1.1 mutation:** strip the record-read step so the loop dispatches blind and the gate MUST FAIL; golden-FALSE per 11.4.201(1) - a loop that converges on attempt 2 with both attempts recorded and no bound exceeded MUST NOT fire it

• **OWED-GATE-074** — CM-STALL-BOUNDED-AND-ESCALATED

- **Must assert:** asserts a declared retry bound exists and that a loop exceeding it without a recorded escalation to a terminal disposition FAILs
- **Blocked on:** depends on the shared attempt record of OWED-GATE-062 landing, plus a declared per-effort-class retry bound that no consumer has supplied
- **Paired 1.1 mutation:** remove the retry bound leaving an unbounded loop and the gate MUST FAIL; record an escalation with no terminal disposition and it MUST FAIL; golden-FALSE per 11.4.201(1) - a loop that converges on attempt 2 with both attempts recorded and no bound exceeded MUST NOT fire it

B. Token-extraction artifacts – not real gate names

These eight tokens are **not** gates. They are artifacts of the ledger's per-line token extractor. They are **deferred rather than removed** because

`gate_ledger_removals.tsv` is only consulted in `check` mode for names **absent** from the ledger — these tokens are still present in the corpus, so a removal citation never fires for them (proven empirically, not assumed). Per §11.4.227(A) a name is never deleted to lower the count; the fix is a **naming/wrap cleanup in the corpus prose**, after which these tokens stop being extracted and the ratchet drops by 8.

Acceptance criteria for every `OWED-NAMING-*` item: the producing prose is rewritten so the token is no longer emitted — range endpoints written with their full `-PROPAGATION` suffix, and hyphen-wrapped names re-flowed so no line ends mid-name — verified by re-running `gate_ledger.sh generate` and confirming the bare token no longer appears in the ledger.

B. Prose-range artifacts (`Constitution.md` line)

The sentence *"Propagation gates CM-COVENANT-114-239 through CM-COVENANT-114-254 + CM-COVENANT-114-257 through CM-COVENANT-114-262"* writes each range endpoint **without** the `-PROPAGATION` suffix the group shares. The extractor stops at the space, emitting a bare token. Each real gate (`CM-COVENANT-114-NNN-PROPAGATION`) is present in all five governance files.

- **OWED-NAMING-003** — `CM-COVENANT-114-239` → real gate `CM-COVENANT-114-239-PROPAGATION`
- **OWED-NAMING-004** — `CM-COVENANT-114-254` → real gate `CM-COVENANT-114-254-PROPAGATION`
- **OWED-NAMING-005** — `CM-COVENANT-114-257` → real gate `CM-COVENANT-114-257-PROPAGATION`
- **OWED-NAMING-006** — `CM-COVENANT-114-262` → real gate `CM-COVENANT-114-262-PROPAGATION`

B. Hyphen line-wrap artifacts (mirror files)

A gate name hyphen-wrapped across a line break yields a prefix ending in `-`, which the extractor's trailing-punctuation stripper (`sed -E 's/[-.,:;)]+$/'`) turns into a bare token. **All four have zero standalone occurrences corpus-wide** (boundary-anchored check, control-needle verified) — they exist only as wrapped prefixes.

- **OWED-NAMING-001** — `CM-AF-KINOPOISK-5-1-DUAL` → real gate `CM-AF-KINOPOISK-5-1-DUAL-COVERAGE` (wrap at `AGENTS.md:960`)
- **OWED-NAMING-002** — `CM-AF-UI-DRIVEN-VIDEO` → real gate `CM-AF-UI-DRIVEN-VIDEO-COVERAGE` (wrap at `AGENTS.md:927`)
- **OWED-NAMING-007** — `CM-OPERATOR-BLOCKED-SELF` → real gate `CM-OPERATOR-BLOCKED-SELF-RESOLUTION-AUDIT` (wrap at `AGENTS.md:306`)
- **OWED-NAMING-008** — `CM-OPERATOR-BLOCKED-SELF-RESOLUTION` → same real gate, wrapped at a different point (`CLAUDE.md:440`)

Note on `CM-COVENANT-114-27-PROPAGATION`

Investigated and **refuted as an artifact**. `Constitution.md:2645` reads "*existing `CM-COVENANT-114-27-PROPAGATION` stays unchanged (literal `11.4.27` unchanged, stays GREEN)*" — a deliberate reference to the real propagation gate for anchor §11.4.27, which exists as a real anchor heading. It is a genuine gate name, not a range truncation, and it falls in the `*-PROPAGATION` scope rather than the mechanism-gate scope covered here.

B. Landed-but-not-wired: the engine-driven propagation gates

Status (§11.4.6 honest, measured 2026-08-20): 29

`cm_covenant_114_*_propagation.sh` gates (§11.4.27, §11.4.234, §11.4.236–§11.4.262) are IMPLEMENTED as executable gate sites driven by `scripts/gates/lib/covenant_propagation_engine.sh` plus a 29-row data pack, each with a paired `*_mutation_test.sh`. Measured facts, not claims:

- `bash -n / sh -n` : 83/83 clean.
- §1.1 mutations: 29×8 fixtures $\rightarrow$ 29 entries, 29 `exit-0`. Every defect class (MISSING / DUPLICATED / DIVERGENT-HEAD / DIVERGENT-BODY / FENCE-MISSING) FAILs; every clean fixture (POINTER-OK / CLEAN control / ANCHOR-BOUNDARY) PASSes.
- Ledger delta, proven on a control scratch tree: BEFORE IMPLEMENTED=36 UNIMPL=449 $\rightarrow$ AFTER IMPLEMENTED=65 UNIMPL=420. The §11.4.227(A) ratchet moved from **violated** ($449 > 420$ fails=1) to **satisfied** ($420 == 420$ fails=0) *because of these files*.

They are NOT yet wired. No runner, hook, or pre-build stage invokes them. Per §11.4.227(A) an anchor's done-state is its SEAM landing, not its TEXT landing; per §11.4.201(8) the ledger's 420 is a metric on a PROXY, and it reaches its true target only when these gates are implemented **and wired and un-holed**. The GREEN ratchet must not be read as "§11.4.227(A) satisfied in substance."

B. Findings from the independent adversarial verification (- -)

Report: `adversarial-review.md` (session scratch). Findings F1–F3 are **defects in the shared engine**, a faithful port of the already-landed 230-template — they therefore exist in `main` today via the 5 pre-existing standalone gates, independent of this batch. These 29 widen the surface; they did not create the hole.

- **OWED-GATE-ADV-F1 (CRITICAL)** — pointer carve-out swallows real propagation loss. All four project-root consumer carriers are `is_pointer_carrier -true` yet restate anchors; deleting a whole anchor block from all four still exits 0 (4 POINTER-INHERITANCE-SKIP, 0 MISSING). The entire consumer layer can lose any anchor with all gates green.

- **OWED-GATE-ADV-F2 (CRITICAL)** — one-line permanent exemption. Adding an `## INHERITED FROM ...` line to a carrier and deleting its block → exit 0. Any carrier can be silently removed from the lockstep set by one documentation-looking line.
- **OWED-GATE-ADV-F3 (CRITICAL)** — lockstep hash truncatable by the corpus's own convention. A compact `- §11.4.N - ...` citation inside a block collapses the hashed region to the heading; 17 of 18 body lines replaced with filler in one carrier → exit 0.
- **OWED-GATE-ADV-F4 (HIGH)** — 29/29 red on the clean tree (55 MISSING each, of which 18 are vendored third-party trees the prune list does not exclude) → a §11.4.201(1) FAIL-bluff surface and §11.4.248 flake-decay risk. §11.4.255/§11.4.256 red is a **real** defect: the canonical constitution mirrors lack the re-minted blocks.
- **OWED-GATE-ADV-F5 (HIGH)** — verdict is a function of cwd; nothing pins the root. From the repo root the scan reaches outside the repository.
- **OWED-GATE-ADV-F6 (HIGH)** — three false-positive refusals of legitimate corpora (sanctioned cross-layer compact summary; §11.4.141 compact index bullet read as DUPLICATED; unrelated EOF appendix marking the last anchor DIVERGENT).
- **OWED-GATE-ADV-F7 (MEDIUM)** — 29 names, zero enforcement (the wiring gap above).
- **OWED-GATE-ADV-F8 (LOW)** — a real `### ... (§11.4.27, ...)` heading the regex misses; DIVERGENT lines print ambiguous basenames.

Acceptance for F1–F6: each fix is a change to the SHARED engine and therefore alters the 5 already-landed gates' behaviour — a §11.4.120 reconciliation, with the golden-FALSE fixture set extended per §11.4.201(1) so the corrected engine still does not refuse a legitimately-compliant corpus. Not attempted in this batch: the operator's implementation lock is in force, and engine surgery under a lock is exactly the unreviewed-change class §11.4.142 forbids.

Acceptance for F7: wire the suite into a runner/pre-build stage, at which point the 420 ledger figure stops being a proxy and starts being enforcement.

C. Merge-introduced deferrals (§ 11.4.263, arrived via upstream c)

Upstream commit `502586c` ("anchor(§11.4.263): process-group signal-safety mandate") named three CM-* gates with neither an implementation nor a registered deferral. The §11.4.227(A) ratchet caught it correctly on the post-merge re-run — `unimplemented=423 baseline=420 fails=1` — which is the mechanism working, not a false positive.

Remediation taken: register the three against tracked items (the sanctioned §11.4.227(A) path). The baseline was NOT raised: raising it is the ratchet-gaming channel the anchor explicitly closes, and these gates remain OWED, not satisfied. Post-registration the canonical wrapper returns `unimplemented=420 baseline=420 fails=0`.

- **OWED-GATE-059** — `CM-COVENANT-114-263-PROPAGATION`. Asserts the §11.4.263 anchor block-start is present exactly once per governance file and lockstep-identical across the mirror set. **Blocked on:** the shared propagation engine carries OWED-GATE-ADV-F1..F3 (Section B); landing a 30th gate on a holed engine widens the surface again. Reconcile the engine first per §11.4.120, then land this gate.
- **OWED-GATE-060** — `CM-KILLPG-PGID-GUARD`. Asserts every `killpg / kill(-pid)` call site in production code carries the `pid > 1 && pgid > 1` guard. **Blocked on:** needs a per-language call-site scanner plus its paired §1.1 mutation (remove the guard → the regression test must FAIL, per §11.4.115 observation-before-trust).
- **OWED-GATE-061** — `CM-TEST-MOCK-PID-EXPLICIT-INT`. Asserts every subprocess mock in test code sets `mock.pid` explicitly as `int` with `killpg` patched. **Blocked on:** needs a test-corpus scanner plus its paired §1.1 mutation (unset `mock.pid` → the assertion must catch the `pgid <= 1` attempt).

Honest boundary (§11.4.6): a DEFERRED gate is owed work that is tracked and visible — it is NOT a gate that runs. §11.4.263's signal-safety mandate is therefore **unenforced mechanically** in this repository today; only its anchor text binds.